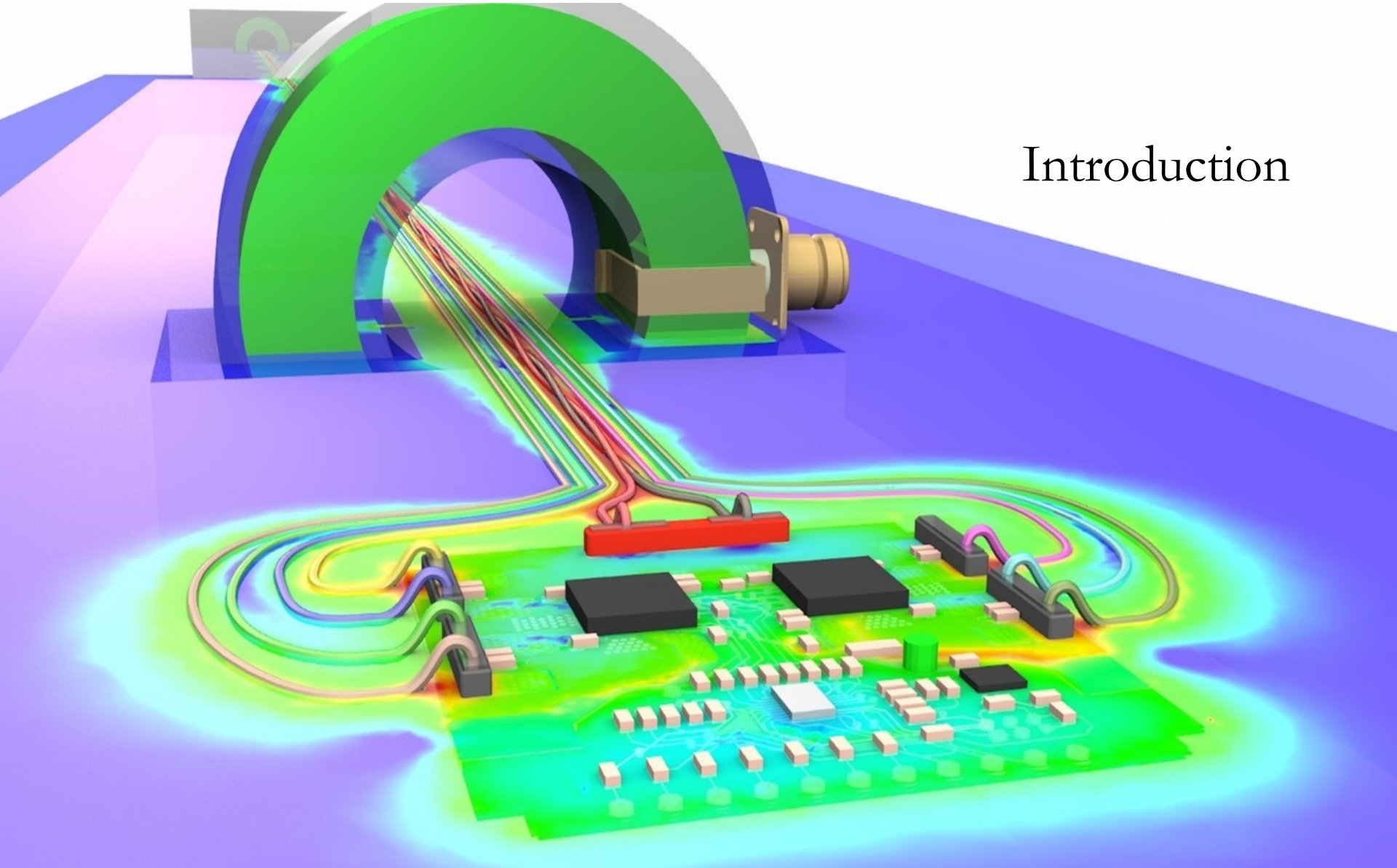
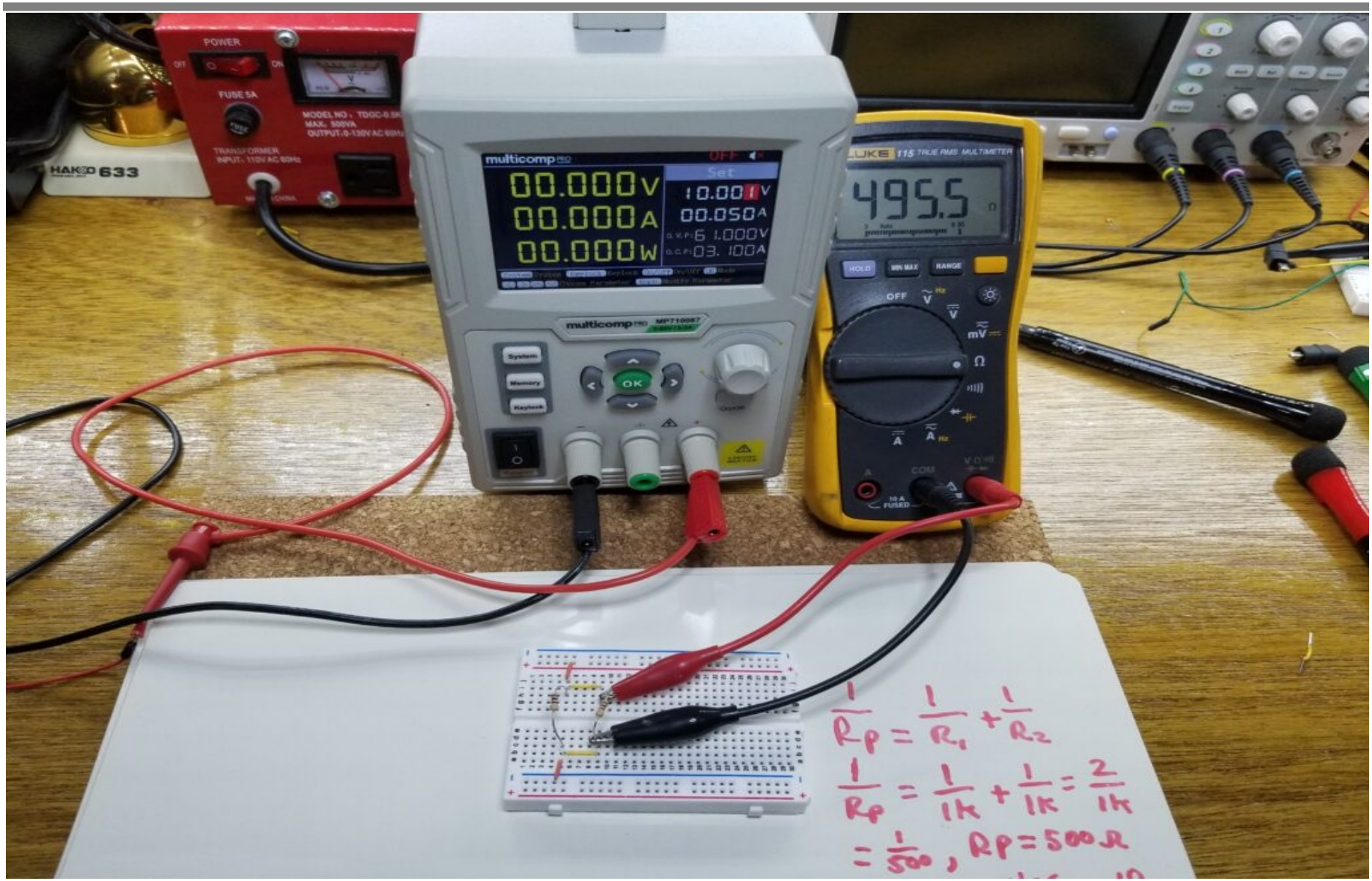
Introduction



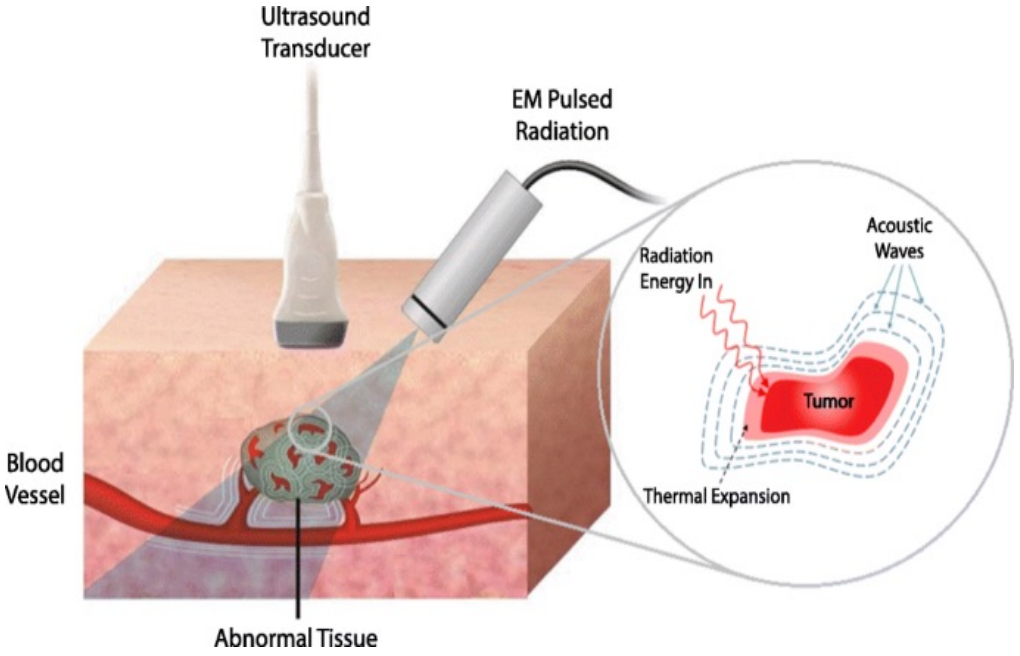
What's Wrong with Circuit Theory?

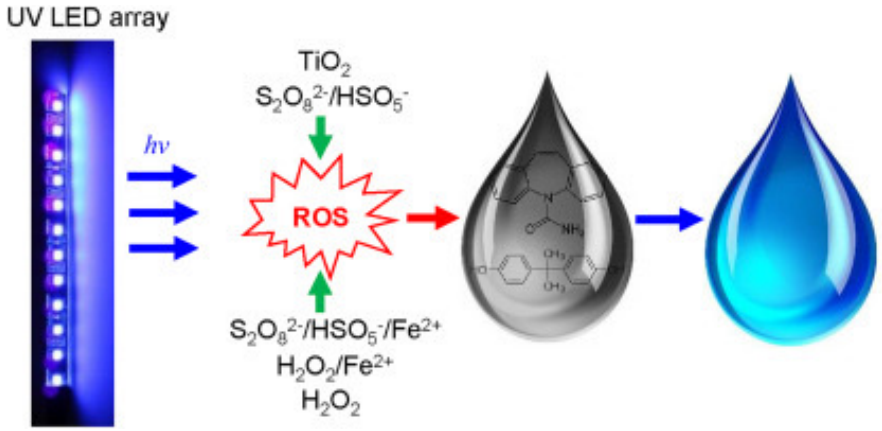


Electromagnetics: Sensing

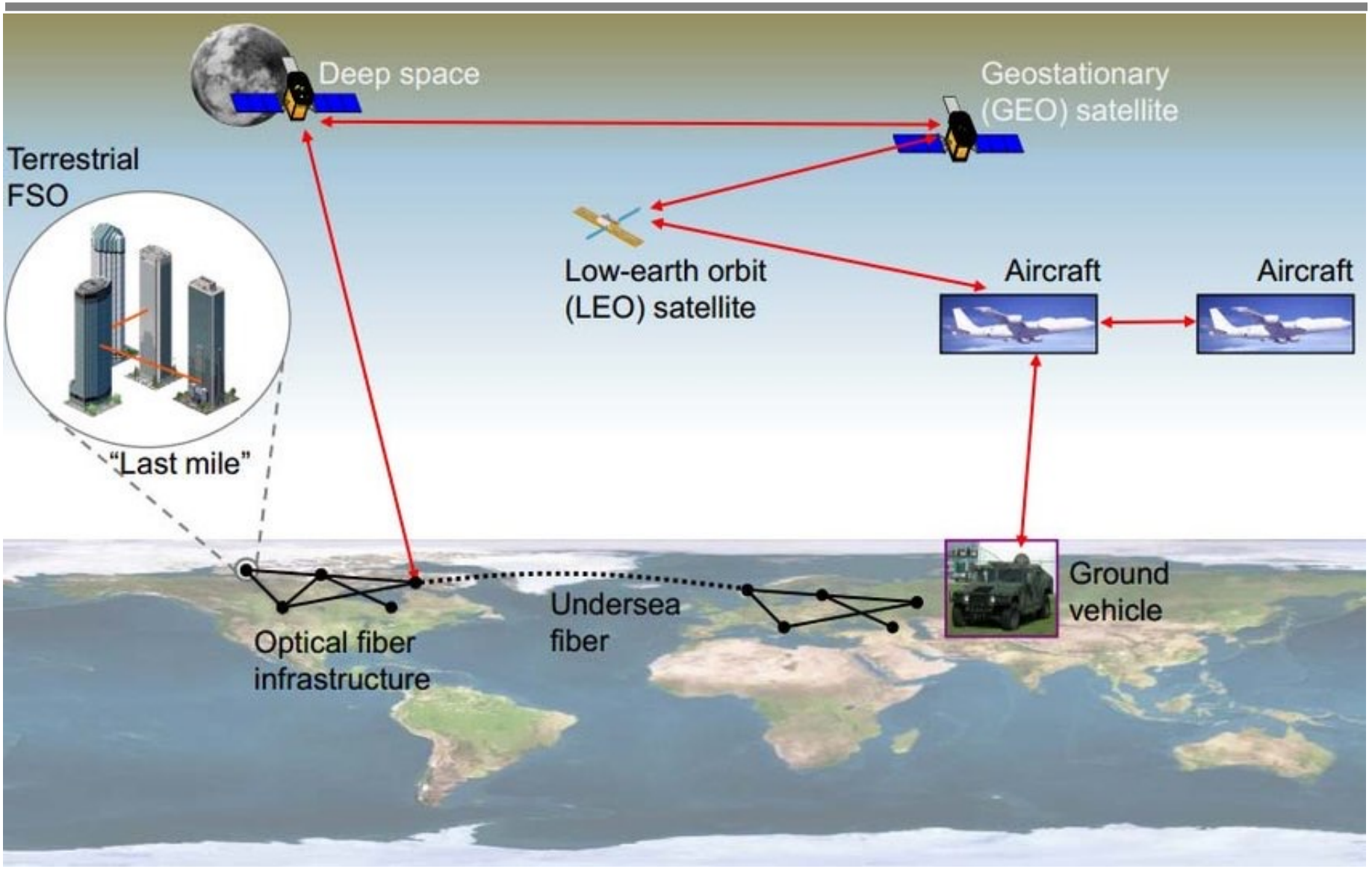








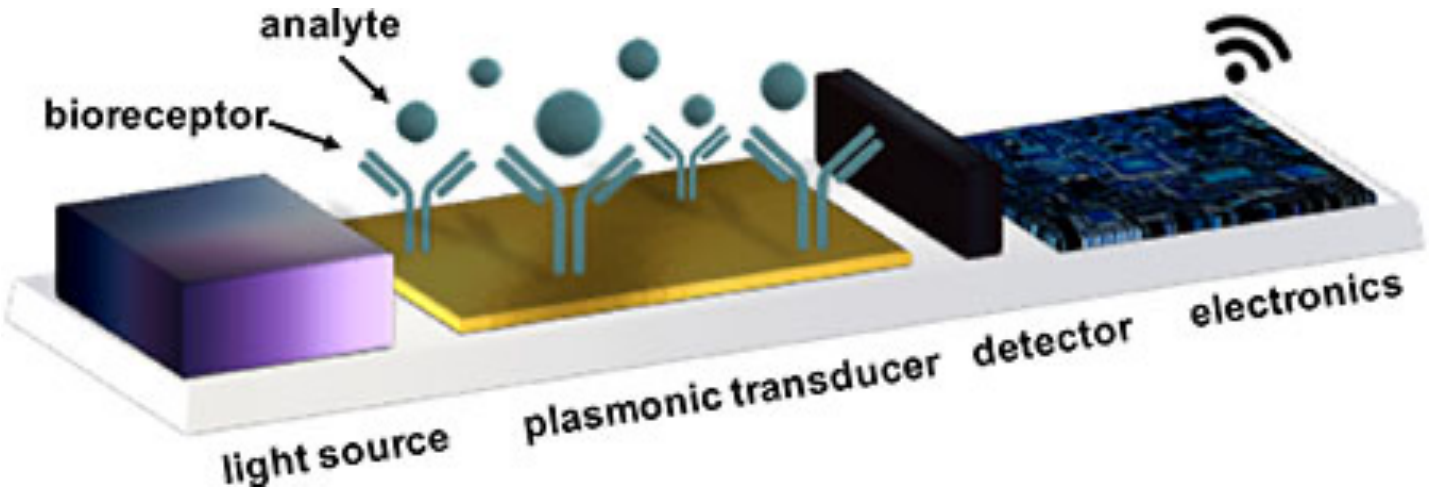
Electromagnetics: Communications

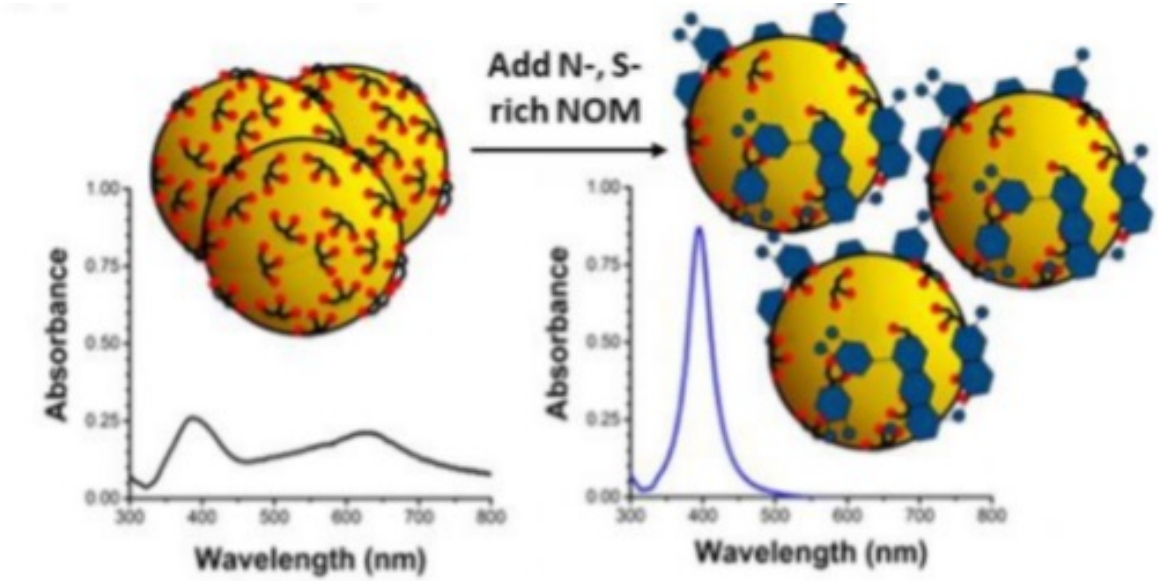


Electromagnetics: Remote Control



Electromagnetics: Bio-medical

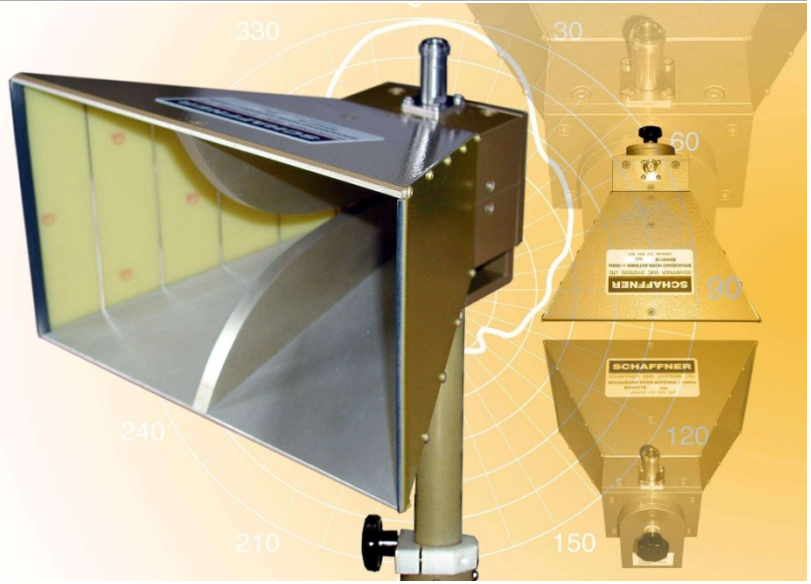


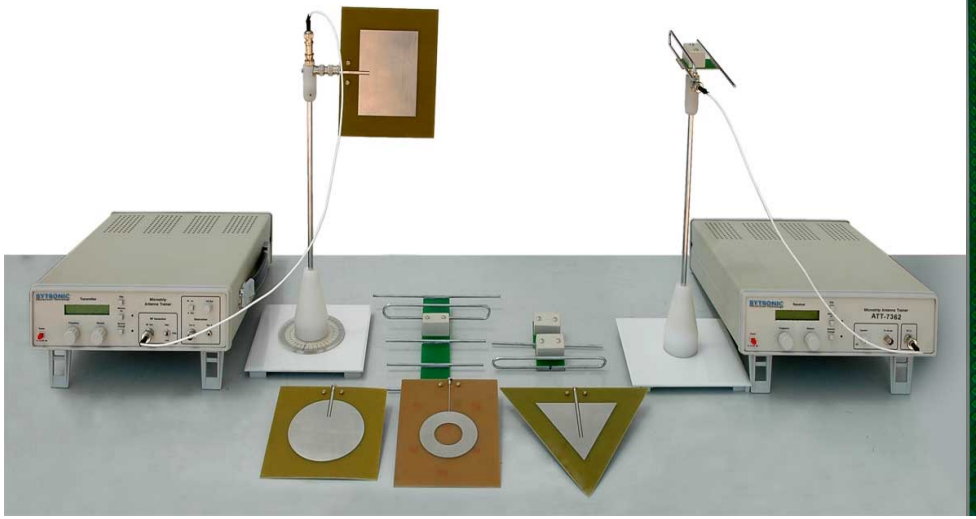


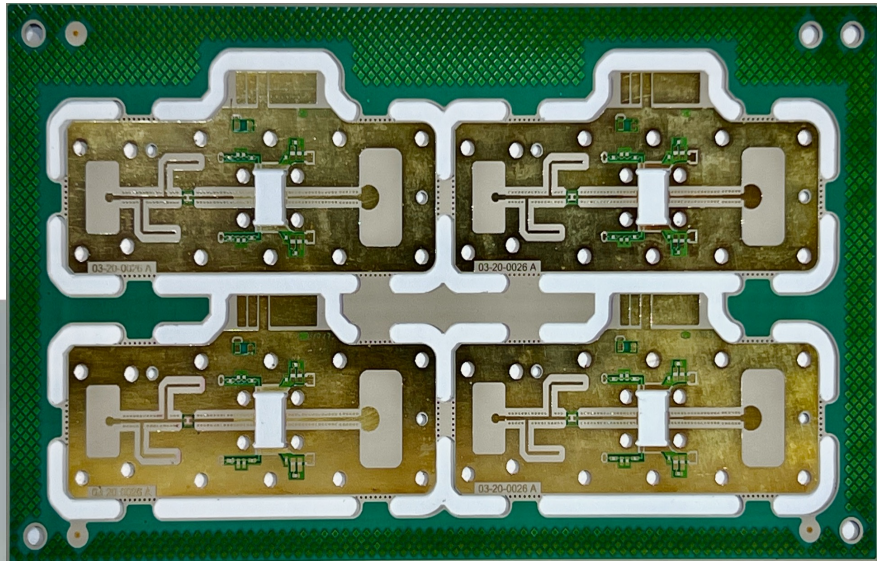
Electromagnetics: Microwave

Nano Photonics and Plasmonics Research









What's Wrong with Circuit Theory?

$$J_{displacement} = \frac{\partial D}{\partial t}$$

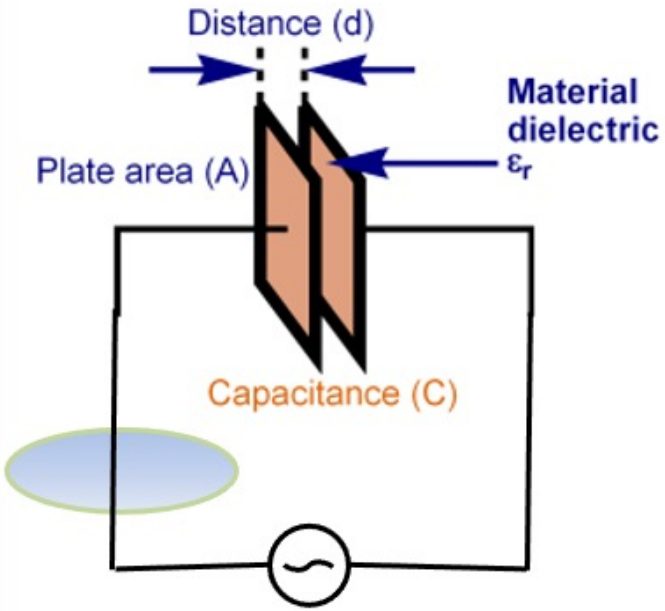
$$i_{displacement} = \int \frac{\partial D}{\partial t}\, dS = \frac{\varepsilon A}{d}\frac{\partial v}{\partial t} = C\frac{\partial v}{\partial t}$$

$$v(t) = V_o \sin(\omega t)$$

$$i_{displacement} = C\omega V_o \cos(\omega t)$$

$$i_{enclosed} = \int H \,.\, dl = 2\pi r H$$

$$H = \frac{CV_o}{2\pi}\frac{\omega}{r} V_o \cos(\omega t)$$



$$i(t) = \frac{\mathrm{d}q(t)}{\mathrm{d}t} = C\frac{\mathrm{d}v(t)}{\mathrm{d}t}$$

Nano Photonics and Plasmonics Research

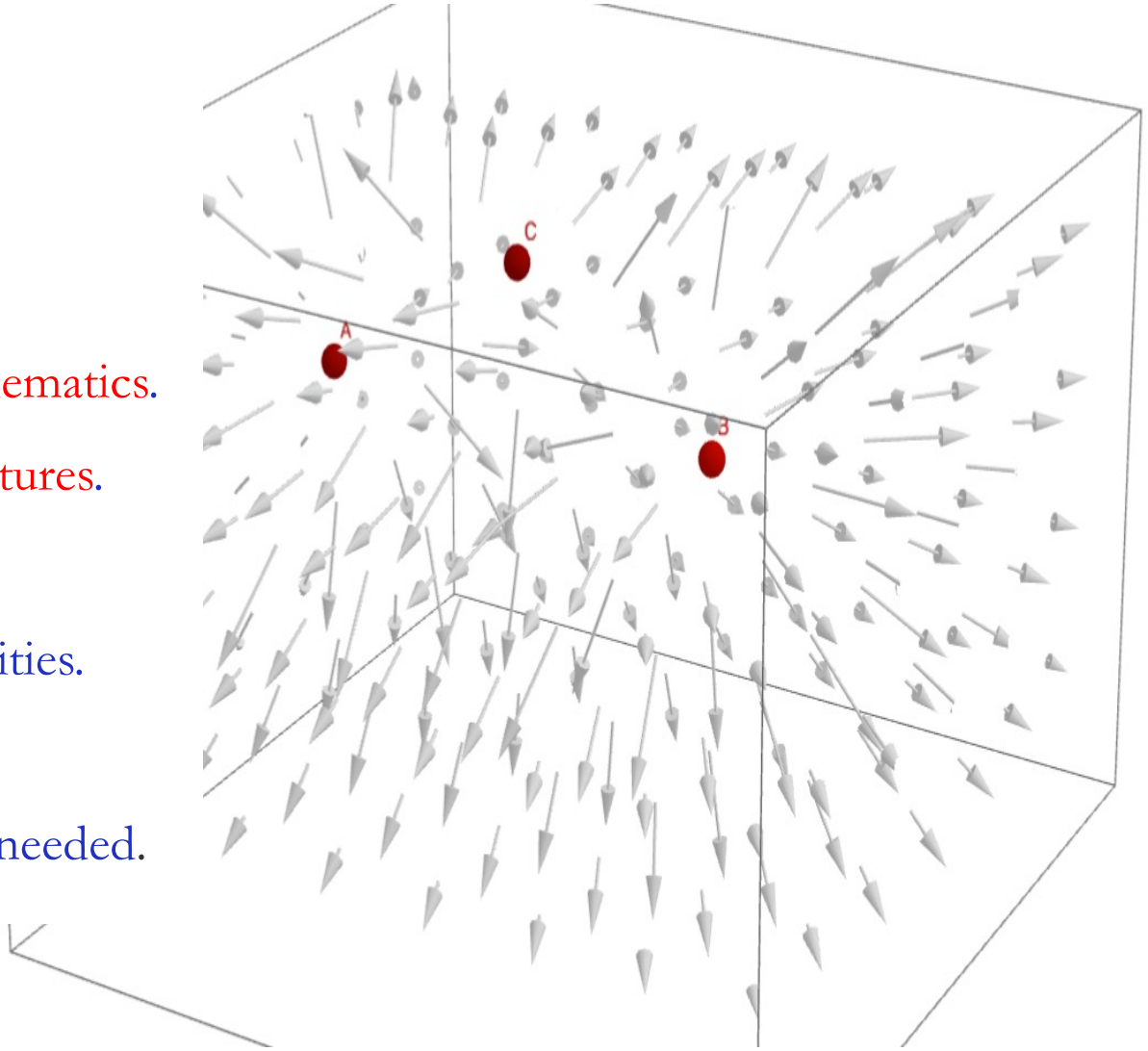
TABLE 1-2
Relations between electromagnetic field and circuit theories

Field theory	Circuit theory
1. $\mathscr{E}$ (electric field intensity)	1. v (voltage)
2. $\mathscr{H}$ (magnetic field intensity)	2. i (current)
3. $\mathscr{D}$ (electric flux density)	3. ϱ_{ev} (electric charge density)
4. $\mathscr{B}$ (magnetic flux density)	4. ϱ_{mv} (magnetic charge density)
5. $\mathscr{J}$ (electric current density)	5. i_e (electric current)
6. $\mathscr{M}$ (magnetic current density)	6. i_m (magnetic current)
7. $\mathscr{J}_d = \varepsilon \frac{\partial \mathscr{E}}{\partial t}$ (electric displacement current density)	7. $i = C\frac{dv}{dt}$ (current through a capacitor)
8. $\mathscr{M}_d = \mu \frac{\partial \mathscr{H}}{\partial t}$ (magnetic displacement current density)	8. $v = L\frac{di}{dt}$ (voltage across an inductor)
9. *Constitutive relations*	9. *Element laws*
(a) $\mathscr{J}_c = \sigma \mathscr{E}$ (electric conduction current density)	(a) $i = Gv = \frac{1}{R}v$ (Ohm's law)
(b) $\mathscr{D} = \varepsilon \mathscr{E}$ (dielectric material)	(b) $\mathscr{Q}_e = Cv$ (charge in a capacitor)
(c) $\mathscr{B} = \mu \mathscr{H}$ (magnetic material)	(c) $\psi = Li$ (flux of an inductor)
10. $\oint_C \mathscr{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t}\iint_S \mathscr{B} \cdot d\mathbf{s}$ (Maxwell–Faraday equation)	10. $\Sigma v = -L_s \frac{\partial i}{\partial t} \simeq 0$ (Kirchhoff's voltage law)
11. $\oiint_S \mathscr{J}_{ic} \cdot d\mathbf{s} = -\frac{\partial}{\partial t}\iiint_V \varrho_{ev}\, dv = -\frac{\partial \mathscr{Q}_e}{\partial t}$ (continuity equation)	11. $\Sigma i = -\frac{\partial \mathscr{Q}_e}{\partial t} = -C_s \frac{\partial v}{\partial t} \simeq 0$ (Kirchhoff's current law)
12. *Power and energy densities*	12. *Power and energy*
(a) $\oiint_S (\mathscr{E} \times \mathscr{H}) \cdot d\mathbf{s}$ (instantaneous power)	(a) $p = vi$ (power–voltage–current relation)
(b) $\iiint_V \sigma \mathscr{E}^2\, dv$ (dissipated power)	(b) $p_d = Gv^2 = \frac{1}{R}v^2$ (power dissipated in a resistor)
(c) $\frac{1}{2}\iiint_V \varepsilon \mathscr{E}^2\, dv$ (electric stored energy)	(c) $\frac{1}{2}Cv^2$ (energy stored in a capacitor)
(d) $\frac{1}{2}\iiint_V \mu \mathscr{H}^2\, dv$ (magnetic stored energy)	(d) $\frac{1}{2}Li^2$ (energy stored in an inductor)

EM Field Theory

EM Fields are:

- Vector quantities.
- Multi-dimensional.
- Require "special" mathematics.
- Complex real-life structures.
- Interact with material.
- Usually complex quantities.
- Static or Time-varying.
- Simulation software is needed.



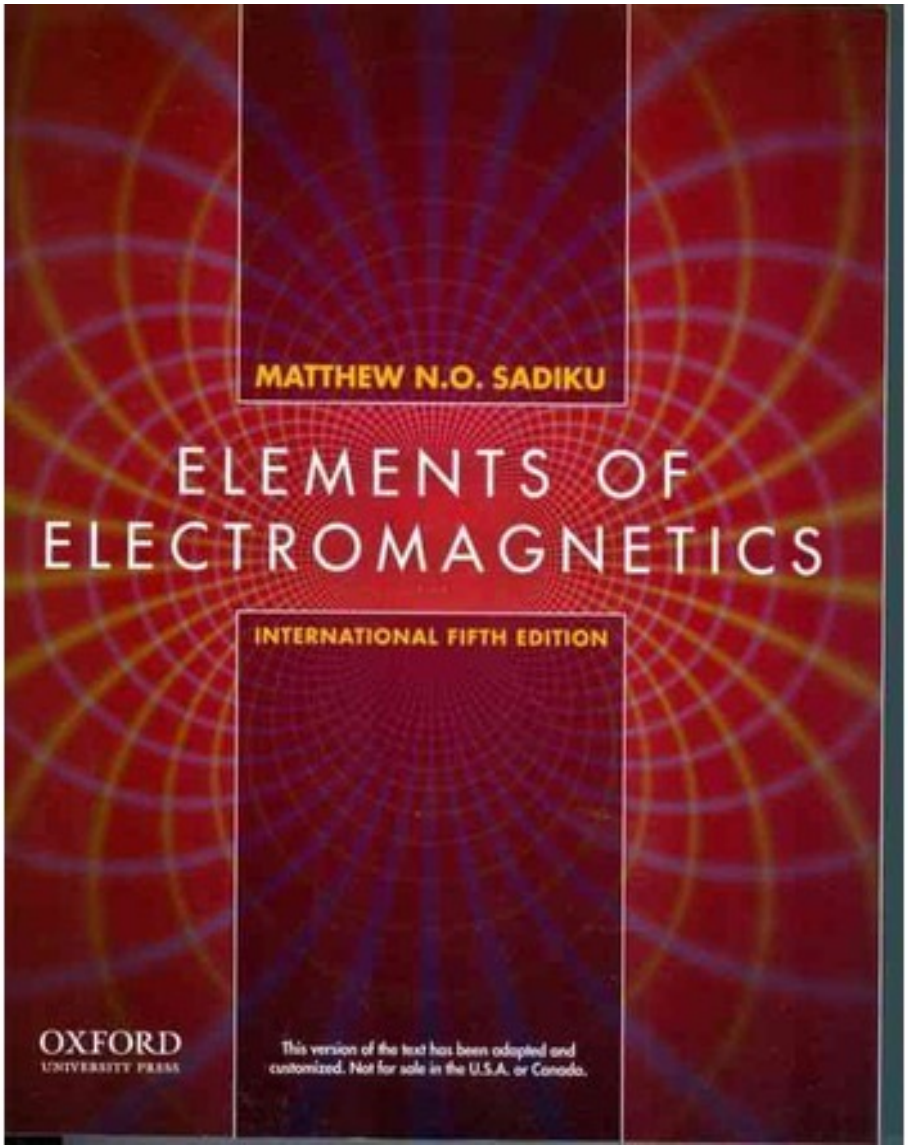
Syllabus

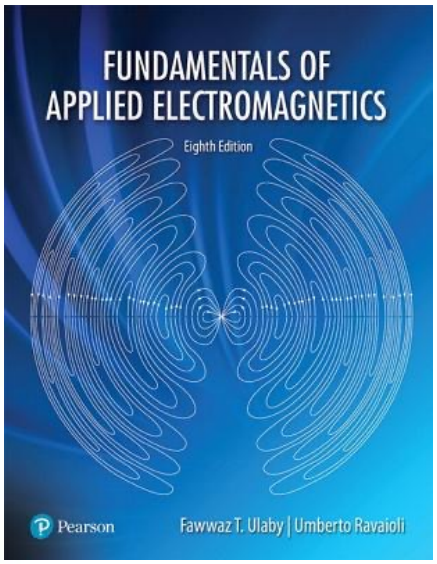
Nano Photonics and Plasmonics Research

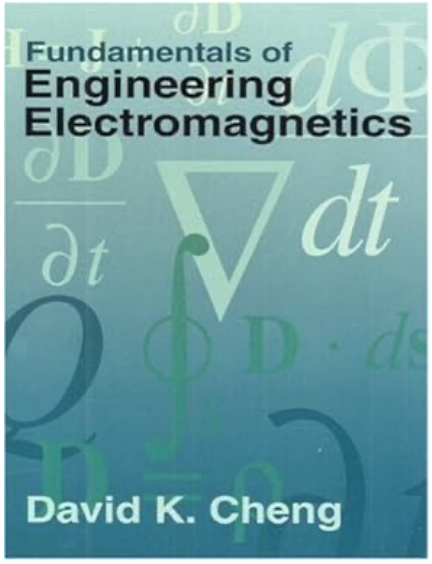
Instructor Information	Mohammad Alsunaidi	Office M4-232	Office Hours Mon-Thu 11:30-12:30	Phone	Email mohammad.alsunaidi@marmara
Course Information	**Textbook**: Elements of Electromagnetics, M. Sadiku, 5th Ed.		Grading	Class Work 30%	Mid-term Exam 30%	Final Exam 40%

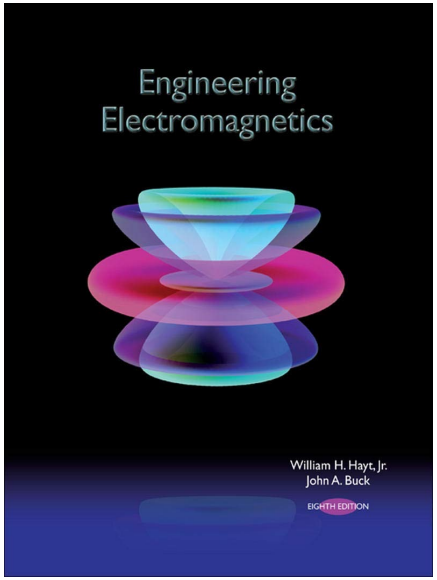
Week	Events	Topics to cover	Ch	Sec	Exercises
1 Oct 2-6	---	**Vectors.** Review of vector analysis and operations on vector. **Coordinate Systems**. Introduction to rectangular, cylindrical, spherical systems.	1 2	1-8 1-3	**(1)** 3, 5, 21, 22 **(2)** 1, 2, 4, 7
2 Oct 9-13	---	**Coordinate Systems**. Representation of points and vectors in all three coordinate systems and their transformations. **Vector Calculus**. Differentials and integrals.	2 3	3-5 1-2	**(2)** 10, 15, 17 **(3)** 4, 5, 7, 10
3 Oct 16-20	Course drop/add Week	**Vector Calculus**. The del operator: Gradient, divergence, curl, and Laplacian operations.	3	3-8	**(3)** 12, 15, 19, 24, 28, 31, 38, 44
4 Oct 23-27	Cumhuriyet Bayramı Oct 29	Coulomb's Law. Gauss' Law. Application of Gauss' Law	4	1-6	(4) 4,5,8,11,16,18,19,20, 22,24
5 Oct 30-Nov 3	Quiz 1 Oct 31 (in-class)	Electric Potential. Electric Dipole, Electrostatic energy density.	4	7-10	(4) 27,29,32,35,38,39,41, 42,43
6 Nov 6-10	Assignment 1 Due Nov 10	Electrostatics in materials: Currents. Polarization, Continuity Equation, Boundary Conditions.	5	1-9	(5) 3,4,10,12,13,15,19, 20,23,25,28,30
7 Nov 13-17	---	Electrostatic Boundary-value Problems: Poisson's & Laplace's Equations (1D). Resistance & Capacitance calculations.	6	1-5	(6) 1,3,6,12,18,20,23, 29,34
Nov 20-26	**Mid-term Exams period**				
9 Nov 27-Dec 1	---	Magnetostatics: Biot-Savart's and Ampere's Laws.	7	1-3	(7) 3,7,10,13
10 Dec 4-8	---	Application of Ampere's Law, Magnetic Flux Density, Magnetic Potentials.	7	4-7	(7) 19,20,23,25
11 Dec 11-15	Assignment 2 Due Dec 15	Magnetic Forces. Magnetization. Magnetic Boundary Conditions.	8	1, 2, 5, 7	(8) 3,5,6,8,9,17,23,26
12 Dec 18-22	Midterm Make-up (Excused, Dec 21-23)	Inductors and Inductance. Magnetic Energy	8	8, 9	(8) 28,31,32
13 Dec 25-29	Quiz 2 Dec 26 (in-class)	Electromagnetics: Faraday's Law and its applications.	9	1-3	(9) 1,3,4,5,7,11
14 Jan 1-5	New Year 2024 Monday	Displacement Current, Maxwell's Equations, Time-harmonic EM Fields	9	4-5, 7	(9) 13,14,15,16,22,28
15 Jan 8-12	Course Project Due Jan 12	**Review**	--	---	---
Jan 15-28	**Final Exams period**	**Jan 29-Feb 4 (Re-sit Final Exams period)**			

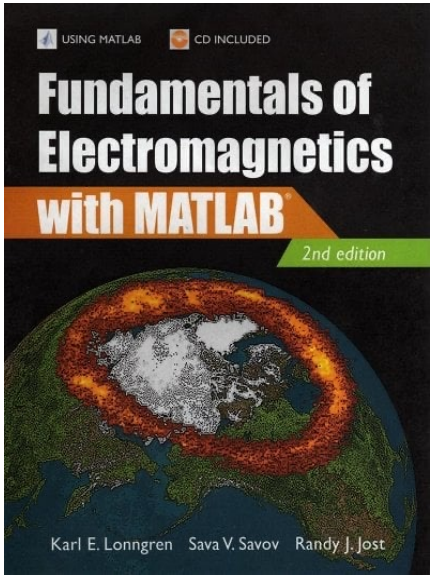
Textbook


MATTHEW N.O. SADIKU
ELEMENTS OF ELECTROMAGNETICS
INTERNATIONAL FIFTH EDITION
OXFORD
UNIVERSITY PRESS
This version of the text has been adapted and customized. Not for sale in the U.S.A. or Canada.

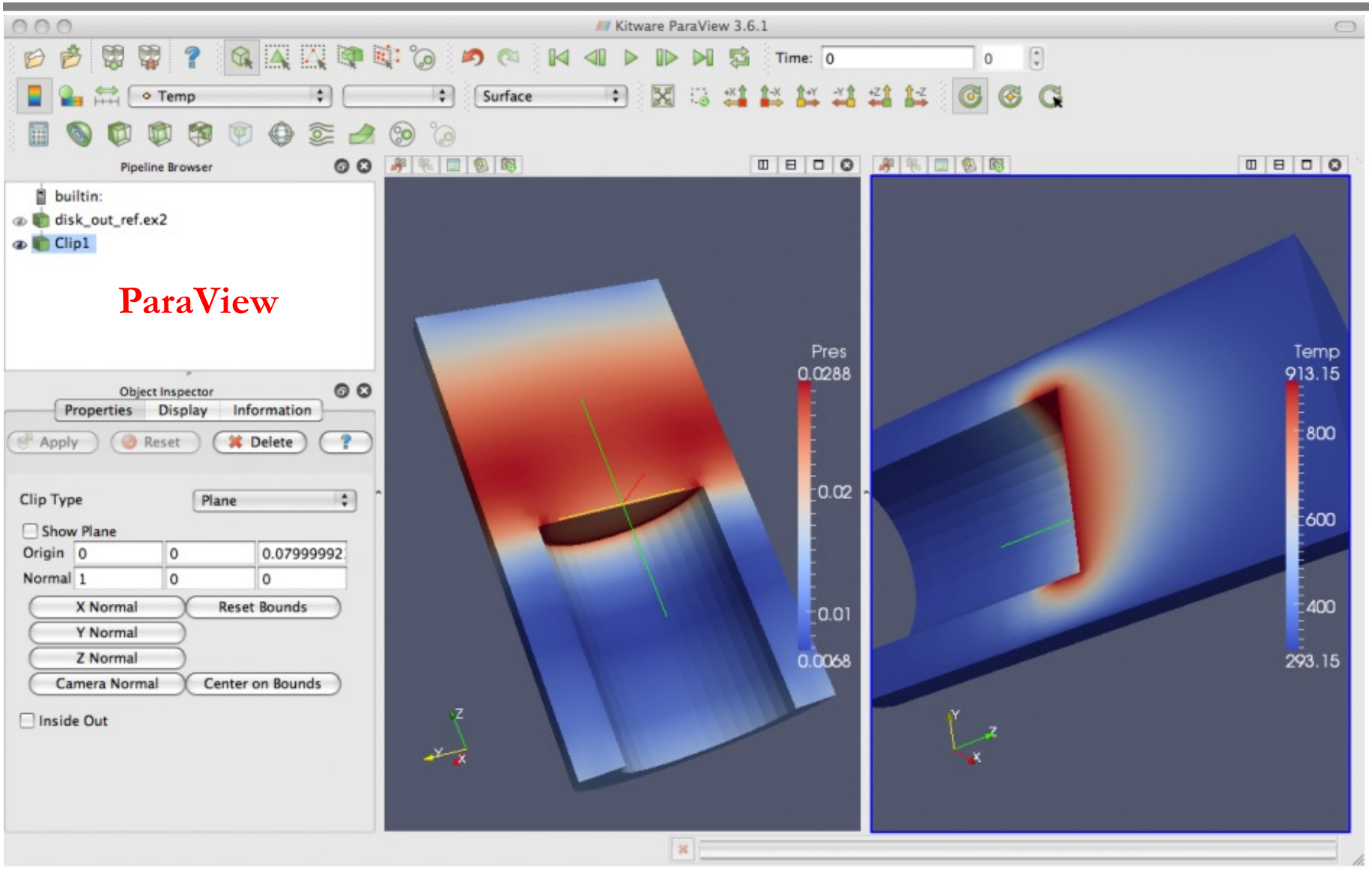

FUNDAMENTALS OF APPLIED ELECTROMAGNETICS
Eighth Edition
Pearson
Fawwaz T. Ulaby | Umberto Ravaioli


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David K. Cheng


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Animation and Visualization Tools



Chapter 1: Vector Analysis

Chapter 2: Coordinate Systems

The following subjects will be discussed

- Vector representation is different systems.
- Transformation between systems.
- Vector operations.
- Examples.

Rectangular Coordinate System

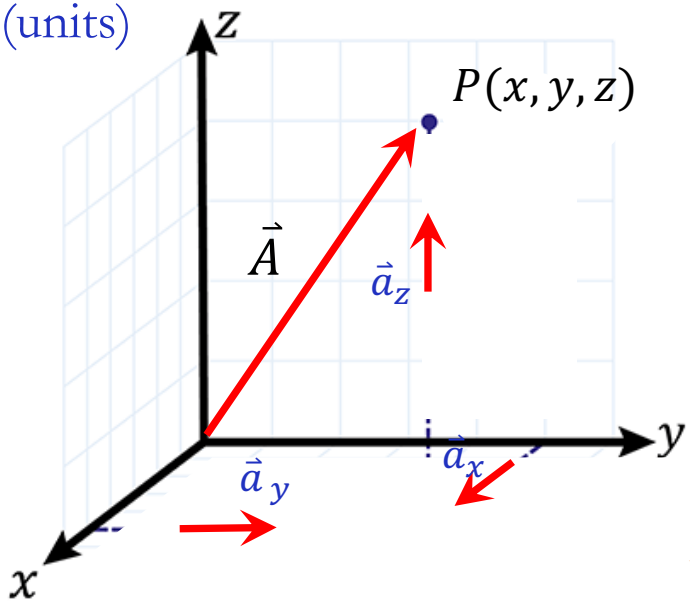
A point in rectangular coordinate system is defined by three lengths.

A vector quantity $\vec{A}$ is generally represented as $\vec{A} = A_x\,\vec{a}_x + A_y\,\vec{a}_y + A_z\,\vec{a}_z$ (units)

The vector Magnitude $|\vec{A}| = \sqrt{{A_x}^2 + {A_y}^2 + {A_z}^2}$ (units)

The direction is given by the "unit vector"

$$\vec{a}_A = \frac{\vec{A}}{|A|} \qquad \vec{A} = |A|\,\vec{a}_A$$



Example

$$\vec{E} = 28\,\vec{a}_x - 13\,\vec{a}_y + 18\,\vec{a}_z \quad \text{(mV/m)}$$

$$|\vec{E}| = \sqrt{1277} \quad \text{(mV/m)}$$

$$\vec{a}_E = \frac{28\,\vec{a}_x - 13\,\vec{a}_y + 18\,\vec{a}_z}{\sqrt{1277}} = 0.78\,\vec{a}_x - 0.36\,\vec{a}_y + 0.5\,\vec{a}_z$$

Cylindrical Coordinate System

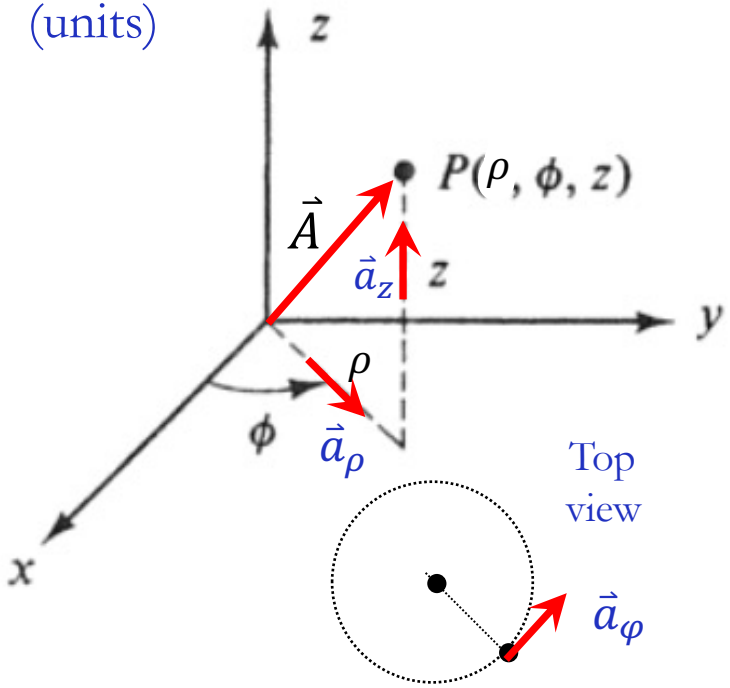
A point in cylindrical coordinate system is defined by two lengths and one angle.

A vector quantity $\vec{A}$ is generally represented as $\vec{A} = A_\rho\,\vec{a}_\rho + A_\varphi\,\vec{a}_\varphi + A_z\,\vec{a}_z$ (units)

The vector Magnitude $|\vec{A}| = \sqrt{{A_\rho}^2 + {A_\varphi}^2 + {A_z}^2}$ (units)

The direction is given by the "unit vector"

$$\vec{a}_A = \frac{\vec{A}}{|A|} \qquad \vec{A} = |A|\,\vec{a}_A$$



Example

$$\vec{H} = 3\,\vec{a}_\rho + 7\,\vec{a}_\varphi - \vec{a}_z \qquad \text{(mA/m)}$$

$$|\vec{H}| = \sqrt{59} \qquad \text{(mA/m)}$$

$$\vec{a}_H = \frac{3\,\vec{a}_\rho + 7\,\vec{a}_\varphi - \vec{a}_z}{\sqrt{59}} = 0.39\,\vec{a}_\rho + 0.91\,\vec{a}_\varphi - 0.13\,\vec{a}_z$$

Spherical Coordinate System

A point in spherical coordinate system is defined by one length and two angles.

A vector quantity $\vec{A}$ is generally represented as $\vec{A} = A_r\,\vec{a}_r + A_\theta\,\vec{a}_\theta + A_\varphi\,\vec{a}_\varphi$ (units)

The vector Magnitude $|\vec{A}| = \sqrt{{A_r}^2 + {A_\theta}^2 + {A_\varphi}^2}$ (units)

The direction is given by the "unit vector"

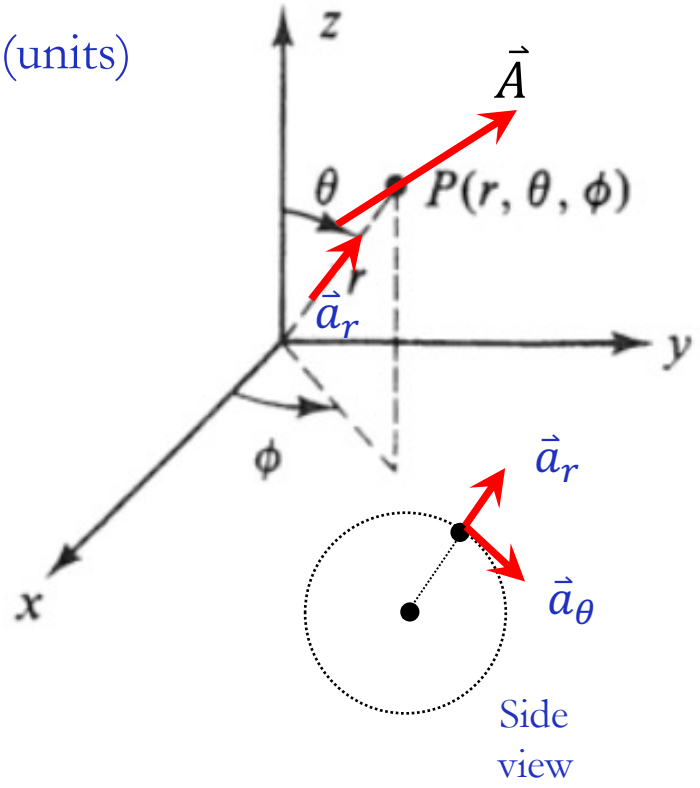
$$\vec{a}_A = \frac{\vec{A}}{|A|} \qquad \vec{A} = |A|\,\vec{a}_A$$

Example

$$\vec{F} = \vec{a}_r - \vec{a}_\theta - \vec{a}_\varphi \qquad \text{(N)}$$

$$|\vec{F}| = \sqrt{3} \qquad \text{(F)}$$

$$\vec{a}_F = \frac{\vec{a}_r - \vec{a}_\theta - \vec{a}_\varphi}{\sqrt{3}}$$



Transformation Between Systems

Cylindrical to Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Rectangular to Cylindrical

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Spherical to Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Spherical to Cylindrical

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Point transformation

$$\rho = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}, \qquad z = z$$

$$x = \rho\cos\phi, \qquad y = \rho\sin\phi, \qquad z = z$$

$$r = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}, \qquad \phi = \tan^{-1}\frac{y}{x}$$

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta$$

Example

$P(-2, 6, 3)$

$\rightarrow P(6.23, 108.43°, 3)$

$\rightarrow P(7, 64.62°, 108.43°)$

Transformation Between Systems

Example: Express the vector $\vec{A} = 3\cos\varphi\ \vec{a}_\rho - 2\rho\ \vec{a}_\varphi + 5\ \vec{a}_z$ in rectangular coordinates.

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3\cos\varphi \\ -2\rho \\ 5 \end{bmatrix}$$

$$\vec{A} = (3\cos^2\varphi + 2\rho\sin\varphi)\ \vec{a}_x + (3\sin\varphi\cos\varphi - 2\rho\cos\varphi)\ \vec{a}_y + 5\ \vec{a}_z$$

$$\rho = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}, \qquad z = z$$

$$x = \rho\cos\phi, \qquad y = \rho\sin\phi, \qquad z = z$$

$$\vec{A} = \left(\frac{3x^2}{x^2+y^2} + 2y\right)\ \vec{a}_x + \left(\frac{3xy}{x^2+y^2} - 2x\right)\vec{a}_y + 5\ \vec{a}_z$$

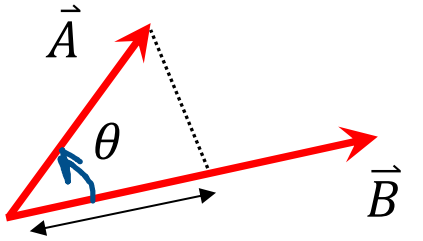
Basic Vector Operations

Scalar Multiplication (dot product)

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

$$\vec{A} \cdot \vec{B} = (A_x \cdot B_x) + (A_y \cdot B_y) + (A_z \cdot B_z)$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$ Scalar result



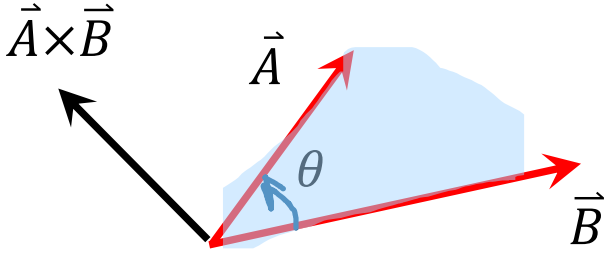
Shadow (projection) of $\vec{A}$ on $\vec{B}$

$|\vec{A}|\cos\theta$

Vector Multiplication (cross product)

$$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\ \vec{a}_n$$ Vector result

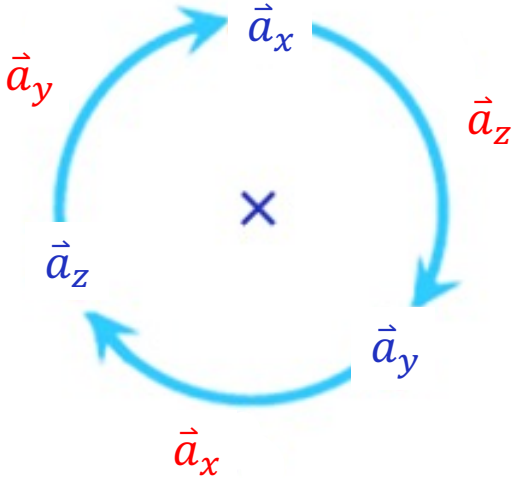
$$\vec{A}\times\vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$



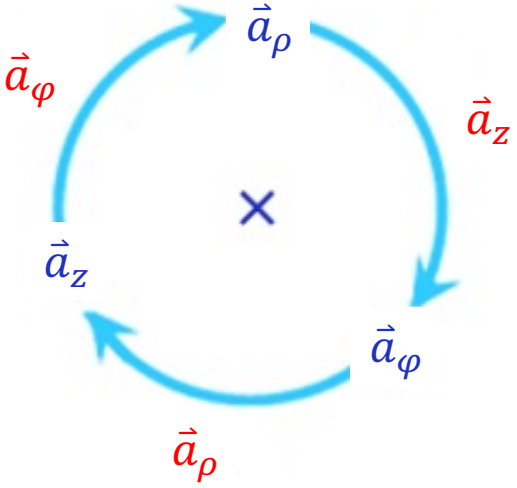
$$\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\vec{a}_x - (A_xB_z - A_zB_x)\vec{a}_y + (A_xB_y - A_yB_x)\vec{a}_z$$

$$\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$$

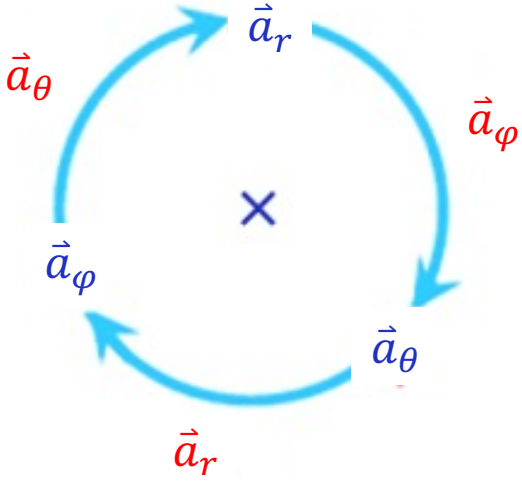
Basic Vector Operations



Rectangular



Cylindrical



Spherical

Basic Vector Operations

Example

Given $\vec{A} = 3\,\vec{a}_x + 4\,\vec{a}_y + \vec{a}_z$ $\vec{B} = 2\,\vec{a}_y - 5\,\vec{a}_z$

Find θ_{AB}.

$$\vec{A} \cdot \vec{B} = (3)(0) + (4)(2) + (1)(-5)$$

$$\vec{A} \cdot \vec{B} = 3$$

$$|\vec{A}| = \sqrt{26} \qquad |\vec{B}| = \sqrt{29}$$

$$\cos\theta_{AB} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} \qquad \theta_{AB} = 83.7°$$

Basic Vector Operations

Example

$$\vec{A} = -4\,\vec{a}_x - 1\,\vec{a}_y + 6\,\vec{a}_z$$

$$\vec{B} = -8\,\vec{a}_x - 2\,\vec{a}_y + 12\,\vec{a}_z$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ -4 & -1 & 6 \\ -8 & -2 & 12 \end{vmatrix} = 0$$

Example

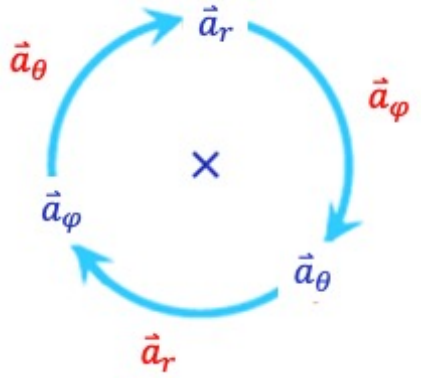
$$\vec{C} = r\,\vec{a}_\theta + \cos\theta\ \vec{a}_\varphi$$

$$\vec{D} = \vec{a}_r - 6\,\vec{a}_\theta + 2\,\vec{a}_\varphi$$

$$\vec{D} \times \vec{C} = (\vec{a}_r - 6\,\vec{a}_\theta + 2\,\vec{a}_\varphi) \times (r\,\vec{a}_\theta + \cos\theta\ \vec{a}_\varphi)$$

$$= r\,\vec{a}_\varphi - \cos\theta\,\vec{a}_\theta - 6r\cos\theta\,\vec{a}_r - 2r\vec{a}_r$$

$$= -r\,(6\cos\theta + 2)\,\vec{a}_r + r\,\vec{a}_\varphi - \cos\theta\,\vec{a}_\theta$$



Examples

Given the electric field $\vec{E} = r^2\,\vec{a}_r + \frac{2}{r}\sin\varphi\,\vec{a}_\theta + \cos\theta\;\vec{a}_\varphi$ (V/m).

Find the magnitude of the field at P$(1, \pi, \pi)$.
Find the magnitude of the field along the $\vec{a}_z$ direction.
At what point(s) does the field vanish?
The component of the field tangential to the spherical surface $r = 1$ at P$(1, \frac{\pi}{2}, \frac{\pi}{2})$.

Q & A

Send me an email if you have any questions.